

## DUSTBADGE · DESIGN PRECIS

# DustBadge design precis

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## DOCUMENT CONTROL

|             |                                        |         |                |
|-------------|----------------------------------------|---------|----------------|
| Document ID | DBG-PRC-001                            | Version | 0.6            |
| Status      | <b>Draft</b>                           | Date    | 2026-09-30     |
| Author      | Amish Chadha                           | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept          |         |                |
| Repository  | github.com/BoujeeEnjinia1701/dustbadge |         |                |
| Source      | docs/02-concept.md @ uncommitted       |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                                                                                  |
|---------|------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.6     | 2026-09-30 | Amish Chadha | Design made constructable (DBG-DDR-003): fixings, gasket, seals and module positions; mass 119.9 g; build plan DBG-BLD-001 linked                                                       |
| 0.5     | 2026-09-26 | Amish Chadha | Budget approved by Amish; parts within the \$91 budget                                                                                                                                  |
| 0.4     | 2026-09-25 | Amish Chadha | Recommendations accepted by Amish (DDR-002): 2,000 mAh cell, R2 range accepted with over-range flag, R3 reference rule, sun use rule; numbers from DBG-CAL-001 v0.2; DBG-DWG-001 Rev P2 |
| 0.3     | 2026-09-25 | Amish Chadha | TRL 3, design choices adopted per DBG-DDR-001; numbers checked against DBG-CAL-001; humidity sensor and thermistor charger added; general arrangement DBG-DWG-001                       |
| 0.2     | 2026-09-25 | Amish Chadha | Populate to TRL 2 (architecture, first-order numbers, safety, media)                                                                                                                    |
| 0.1     | 2026-09-25 | Amish Chadha | Initial scaffold                                                                                                                                                                        |

# DustBadge design precis

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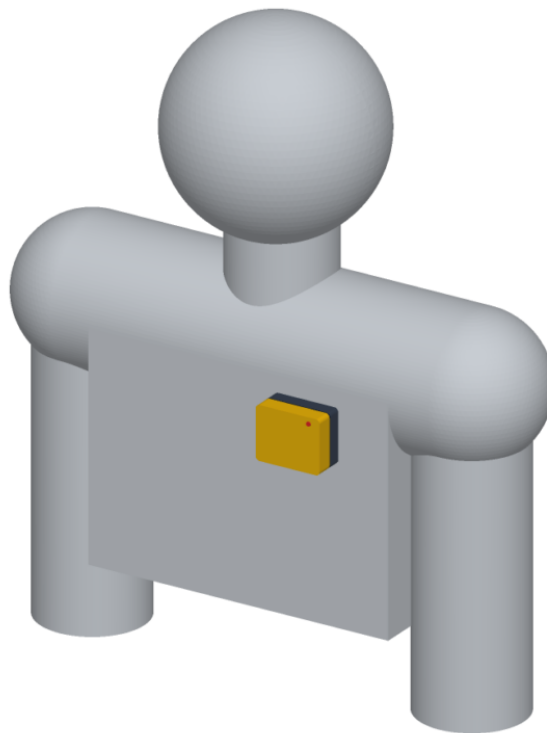
## ■ Summary

DustBadge is a chest-worn badge, 64 x 52 x 30 mm (33 mm with the clip) and about 120 g, that samples respirable dust continuously with an optical particle sensor, converts it to an estimate of respirable crystalline silica using a site calibration, keeps a running 8 h time-weighted average, and vibrates when the projected shift average is heading over the action level. The TRL 3 calculations (DBG-CAL-001 v0.2) show one off-the-shelf sensor, a Bluetooth module and a 2,000 mAh cell run a 12 h shift even at the sensor's maximum current and 0 °C (12.8 h; 17.7 h typical), for \$91 in parts, within the \$91 budget Amish approved on 2026-09-26 (DBG-DDR-002). It cannot identify silica or replace compliance sampling. On paper it misses the hazardous-atmosphere requirement (R14); accuracy at low dust levels (R3) and operation in hot sun (R11) are at risk. The design choices below were decided by Amish on 2026-09-25 (DBG-DDR-002).

### DustBadge

[github.com/BoujeeEnjinia1701/dustbadge](https://github.com/BoujeeEnjinia1701/dustbadge)

CONCEPT, NOT FOR FABRICATION

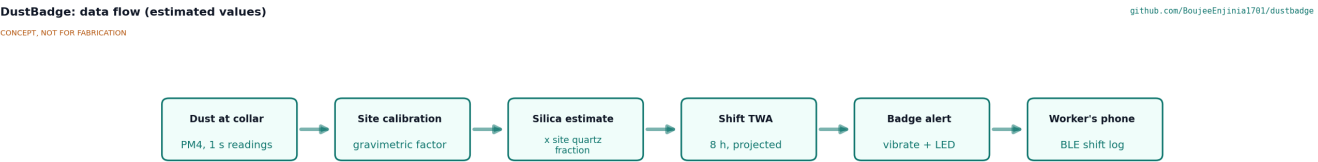


Seen from the front right and above, 24 deg elevation. Grey: Wearer's upper body for scale

## ■ How it works

1. **Sample.** A small fan in the particle sensor draws air in through a downward-facing screened inlet on the bottom of the badge. A laser counts and sizes particles, and the sensor reports mass concentration for particles up to 4  $\mu\text{m}$  (PM4) every second.

2. **Calibrate.** A site factor, taken from at least one co-located filter sample, corrects the optical reading to a gravimetric respirable dust value. NIOSH notes that without such a factor, optical monitors can be off by up to 10 times (Cauda, NIOSH, 2021). Chamber checks against a reference on the lab's CalRig calibration rig catch sensor drift between field samples.
3. **Estimate silica.** Calibrated respirable dust is multiplied by the site silica fraction (the quartz share of respirable dust for that site and task) to give an RCS estimate. With no fraction entered, the badge reports respirable dust only.
4. **Accumulate.** Every minute the firmware logs the 1 min average and updates the running TWA and a projection to the end of the shift at the current rate.
5. **Alert.** The badge vibrates and flashes when the projected 8 h RCS TWA passes the action level, and repeats at the limit. A short high peak (for example a dry cut) gives a separate brief alert so the worker can link the reading to the task.
6. **Sync.** At the end of the shift the worker's phone pulls the log over Bluetooth Low Energy (BLE). The worker decides whether to share it with a supervisor, hygienist or worker organization.



## ■ Main components

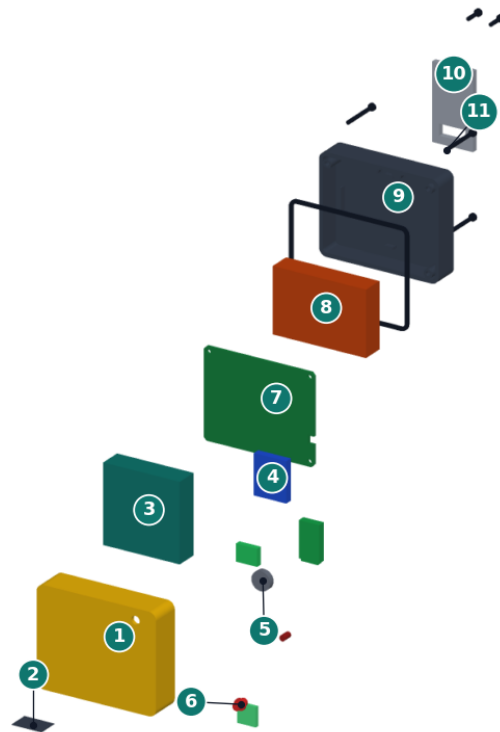
Table 1. Main components, numbered to match the exploded view and bom/bom.csv.

| #  | Component          | Proposed choice                                                                                                                                                                                                                                                         | Notes                                                                                                                        |
|----|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| 1  | Front shell        | 3D-printed PETG, high-visibility yellow, with bottom inlet and outlet slots, USB-C opening and humidity vent; three screw bosses and two sensor stop ribs                                                                                                               | Decided by Amish, 2026-09-25 (DBG-DDR-001, D7)                                                                               |
| 2  | Inlet dust screen  | Stainless mesh, about 1 mm aperture, a 16 x 11 mm square laid inside over the inlet slot and held by the inlet port seal                                                                                                                                                | Keeps grit and splash out; must not act as a size selector for respirable particles                                          |
| 3  | Particle sensor    | Sensirion SPS30 class: PM1, PM2.5, PM4 and PM10 mass; 0 to 1,000 µg/m³; 41 x 41 x 12 mm; 26 g; 45 to 55 mA typical, 65 mA maximum at 5 V; PM4 precision ±25 µg/m³ below 100 µg/m³ and ±25 % above ( <a href="#">datasheet v2.0</a> )                                    | Chosen for its PM4 output. Decided by Amish, 2026-09-25 (DBG-DDR-001, D1)                                                    |
| 4  | Controller and BLE | nRF52840 module with 2 MB flash and LiPo charger (Seeed XIAO nRF52840 class)                                                                                                                                                                                            | Same module family as TremorTrace                                                                                            |
| 5  | Vibration motor    | 10 mm coin motor, stuck to the inside of the front face                                                                                                                                                                                                                 | Felt through clothing when noise makes a beep useless                                                                        |
| 6  | Alert LED          | Red 3 mm LED in a printed clear light pipe with an inside flange, through the front face                                                                                                                                                                                | Visible to the worker looking down and to coworkers                                                                          |
| 7  | Carrier board      | Perfboard carrying the controller and a 500 mA charger module with USB-C and a cell thermistor input; a 5 V boost module for the sensor and an SHT4x-class humidity and temperature breakout at a vent in the bottom face sit on the inside of the front face; polyfuse | Perfboard for the first build; humidity sensor and thermistor added at TRL 3 (DBG-CAL-001, sections B and C)                 |
| 8  | Battery            | 2,000 mAh 3.7 V protected LiPo, about 50 x 34 x 11.5 mm, 38 g, with thermistor                                                                                                                                                                                          | Decided by Amish, 2026-09-25 (DBG-DDR-002, D8); was 1,500 mAh. Fills the space behind the carrier board with 0.5 mm to spare |
| 9  | Rear shell         | 3D-printed PETG with a flat printed TPU gasket and two TPU sensor port seals; three tubes that press the carrier board onto the front bosses, four cell locating ribs and two clip screw bosses                                                                         | Holds the cell away from the body side; closed by three M2 x 20 screws from the rear face (DBG-DDR-003)                      |
| 10 | Clip               | Stainless spring clip with strap loop, held by two M2 x 8 screws                                                                                                                                                                                                        | For a shirt pocket, collar, harness or hi-vis vest                                                                           |

**DustBadge: exploded view**[github.com/BoujeeEnjinia1701/dustbadge](https://github.com/BoujeeEnjinia1701/dustbadge)

CONCEPT, NOT FOR FABRICATION

- 1 Front shell with bottom inlet
- 2 Inlet dust screen
- 3 Optical particle sensor (SPS30 class)
- 4 Controller and BLE module
- 5 Vibration motor
- 6 Alert LED and light pipe
- 7 Carrier board and its modules
- 8 LiPo cell, 2,000 mAh, protected
- 9 Rear shell, TPU gasket and port seals
- 10 Spring clip and strap loop
- 11 M2 screws (3 shell, 2 clip)

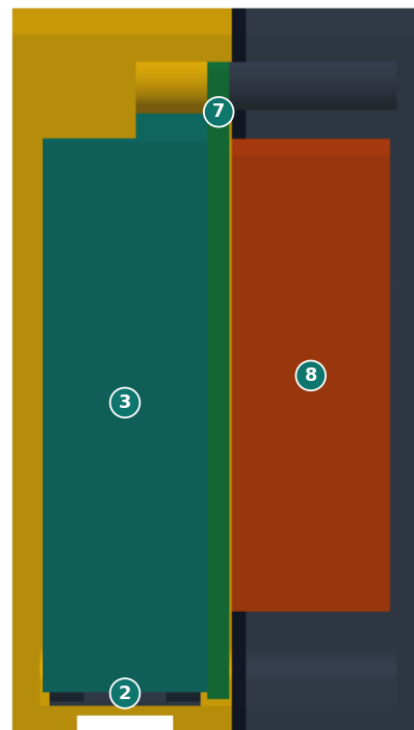


Seen from the front right and above, 24 deg elevation; numbers match bom/bom.csv

**DustBadge: cutaway, section through the particle sensor inlet**[github.com/BoujeeEnjinia1701/dustbadge](https://github.com/BoujeeEnjinia1701/dustbadge)

CONCEPT, NOT FOR FABRICATION

- 2 Inlet dust screen
- 3 Optical particle sensor (SPS30 class)
- 7 Carrier board and its modules
- 8 LiPo cell, 2,000 mAh, protected



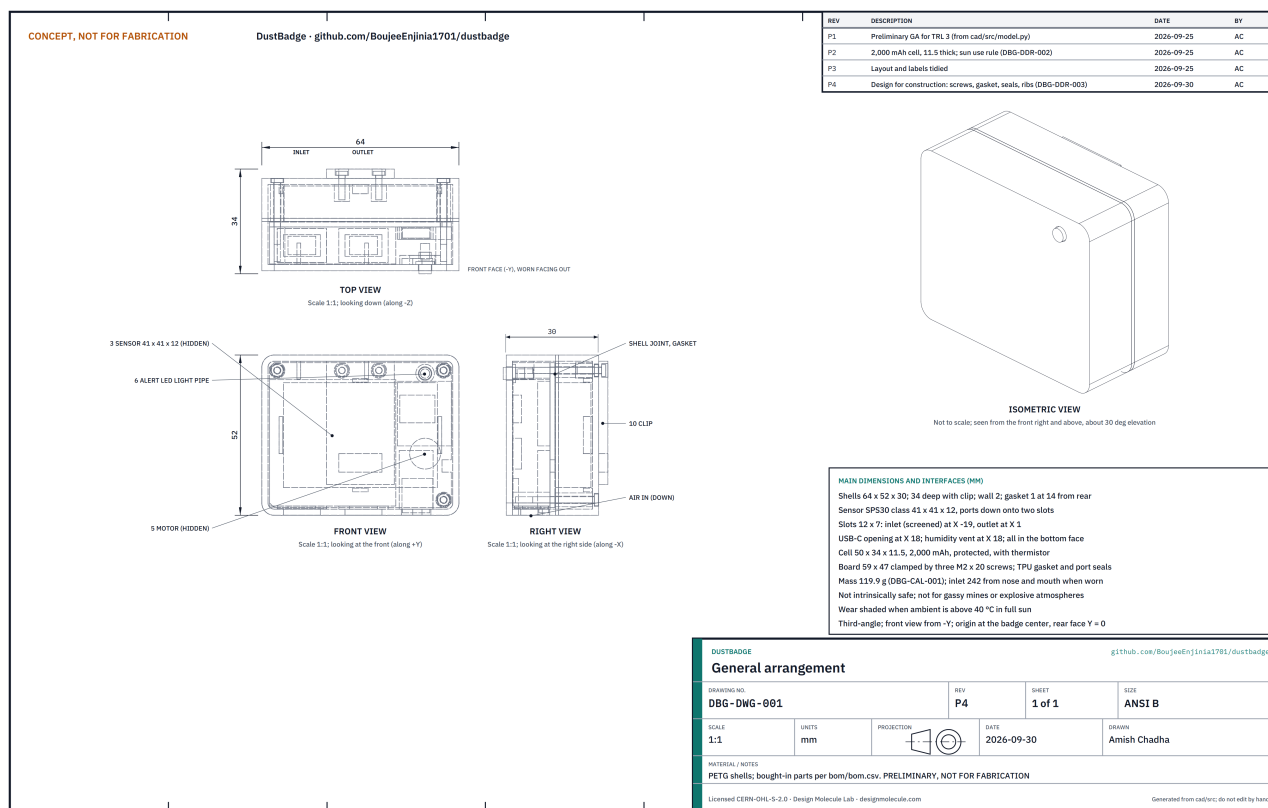
Front of the badge (worn facing out) on the left. Yellow: front shell (1); dark grey: rear shell (9). Air enters upward through the bottom screen (2)

## ■ Numbers checked at TRL 3

All values come from DBG-CAL-001 v0.2 and its script `docs/04-calcs/sizing.py` ; the bracketed tags point to its output lines.

Table 2. Key numbers.

| Quantity                                         | Value                                                                                                         | Basis                                                                                                    | Requirement              |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------------------------|
| Power drawn from the cell                        | 335 mW typical, 393 mW maximum                                                                                | Sensor 55 or 65 mA at 5 V through an 85 % boost, plus 11 mW for controller, BLE and humidity sensor [A3] |                          |
| Usable cell energy                               | 5.92 Wh; 5.03 Wh at 0 °C                                                                                      | 2,000 mAh x 3.7 V, 80 % usable, 85 % at 0 °C [A1]                                                        |                          |
| Run time, continuous sampling                    | 17.7 h typical; 12.8 h at maximum current and 0 °C                                                            | [A3]                                                                                                     | R7 met                   |
| Cell for 12 h in the worst case                  | about 1,880 mAh; the 2,000 mAh cell gives a 7 % margin                                                        | [A4, A6]                                                                                                 |                          |
| Charge time                                      | 4.2 h at 500 mA                                                                                               | [B1]                                                                                                     |                          |
| Shell temperature in full sun                    | about 48 °C at 30 °C ambient, 63 °C at 45 °C ambient; at most 57.8 °C under the use rule (shaded above 40 °C) | Solar and electrical heat against surface losses [C1, C3]                                                | R11 at risk              |
| Respirable dust at the action level              | 250 µg/m³ at 10 % silica; 31 µg/m³ at 80 % silica                                                             | 25 µg/m³ RCS divided by the silica fraction [E2]                                                         | Within the 1 mg/m³ range |
| PM4 precision before site calibration            | ±25 µg/m³ below 100 µg/m³                                                                                     | Datasheet; 80 % of the dust level at the action level for quartz-rich stone [F4]                         | Site factor essential    |
| Upper end of useful range                        | 1 mg/m³ with over-range flag; limit in range for silica fractions of 5 % or more                              | Sensor specification [E1, E3]                                                                            | R2 met as restated       |
| Shift average uncertainty after site calibration | ±21 to ±29 % (quarry); ±22 to ±30 % (quartz-rich stone with the 4.2 L/min, two-shift reference)               | Root sum of squares, assumed terms [F2, F3]                                                              | R3 at risk               |
| Log size                                         | 11.5 kB per 12 h shift; about 182 shifts in 2 MB                                                              | 16-byte record each minute [H1]                                                                          | R12 met                  |
| Inlet to nose and mouth                          | 242 mm                                                                                                        | Model, collar or upper-strap mount [I1]                                                                  | R8 met                   |
| Mass                                             | 119.9 g                                                                                                       | Shells, gasket and seals from model volume, parts from datasheets [J2]                                   | R9 met, 0.1 g margin     |
| Size                                             | 64 x 52 x 30 mm; 33 mm deep with the clip                                                                     | Model                                                                                                    | R9 met                   |
| Parts cost                                       | \$91.00                                                                                                       | bom/bom.csv [L1]                                                                                         | R15 met (\$91 budget)    |



## Key design choices

- **Estimate, never claim, silica.** The badge shows respirable dust as the measured quantity and RCS only as a labeled estimate from a site fraction. Decided by Amish, 2026-09-25 (DBG-DDR-001, D2).
- **PM4 sensor rather than a cheaper PM2.5 sensor.** The SPS30 class reports the size band closest to the respirable convention. A Plantower PMS5003 costs less but reports only up to PM10 with no PM4 bin. Decided by Amish, 2026-09-25 (DBG-DDR-001, D1).
- **Inlet facing down.** Reduces direct deposition of large chips and water spray into the sensor. Decided by Amish, 2026-09-25 (DBG-DDR-001, D7).
- **Continuous sampling.** Cutting and grinding tasks can last a minute or two, so the fan runs all shift rather than duty cycling, which the 2,000 mAh cell covers (R7 met on paper). Decided by Amish, 2026-09-25 (DBG-DDR-001, D4).
- **Default limits.** US OSHA and MSHA values by default (25  $\mu\text{g}/\text{m}^3$  action level, 50  $\mu\text{g}/\text{m}^3$  limit, 8 h TWA), configurable for other jurisdictions such as the EU 100  $\mu\text{g}/\text{m}^3$ . Decided by Amish, 2026-09-25 (DBG-DDR-001, D3).
- **Larger cell.** 2,000 mAh so that R7 holds at the sensor's maximum current and at 0 °C, at 8 g and \$3. Decided by Amish, 2026-09-25 (DBG-DDR-002, D8).
- **Projected TWA alert.** Alerting on the projected shift average (dose so far plus the last 60 min mean rate) rather than on instantaneous readings avoids alarms every time a truck passes, while a separate brief peak alert (1 min mean over 250  $\mu\text{g}/\text{m}^3$  RCS) still links dust to tasks. DBG-CAL-001 section G checks five scenarios.



- **Over-range readings count, not vanish.** A minute above the sensor's 1 mg/m<sup>3</sup> range is logged at the range limit, flagged and counted toward the projection, so the badge alerts rather than going quiet. The 1 mg/m<sup>3</sup> range with this flag is accepted for the first sectors (surface quarries and stone fabrication) instead of a wider-range sensor. Decided by Amish, 2026-09-25 (DBG-DDR-002, D11).
- **Reference sample at low-dust, high-silica sites.** Where respirable dust at the action level is below about 100 µg/m<sup>3</sup>, the site factor comes from a 4.2 L/min cyclone over two shifts, so the filter holds enough mass to weigh. Decided by Amish, 2026-09-25 (DBG-DDR-002, D9).
- **Sun use rule.** Wear the badge shaded, under a vest flap or out of direct sun, when the ambient is above 40 °C in full sun. A lighter shell is a later option, to be tested at TRL 4, which is on hold. Decided by Amish, 2026-09-25 (DBG-DDR-002, D10).
- **Humidity flag.** The added humidity sensor flags readings above 80 % RH, the sensor's best-performance limit, and sudden rises during wet cutting or spraying.
- **Worker-owned data.** Logs stay on the badge and the worker's phone unless the worker shares them, and are never used for discipline. Decided by Amish, 2026-09-25 (DBG-DDR-001, D5).

## ■ Dependencies on other lab projects

- **CalRig** provides the chamber for periodic badge-to-badge and drift checks. Its TRL 2 design uses incense smoke from 5 to 300 µg/m<sup>3</sup> and states that results will not transfer directly to mineral dust, so it cannot characterize the sensor above 1 mg/m<sup>3</sup> or set a site factor. Field gravimetric samples are still needed for the site factor, because chamber aerosol is not site dust. The badge (64 x 52 x 33 mm) fits CalRig's 400 x 300 x 300 mm chamber.
- The badge does not use FieldNode, which is a fixed outdoor node; a FieldNode with the same sensor could log area dust at a site as a later option.

## ■ Safety

**Safety:** DustBadge is a research and educational prototype. It is not certified monitoring equipment or personal protective equipment, and it does not replace regulatory exposure sampling, engineering controls or respirators. A low reading does not mean the air is safe: the sensor cannot see silica directly, can drift, and can under-read outside its calibration.

**Safety:** It is not intrinsically safe. Do not use it in underground coal mines or in any place where flammable gas or combustible dust may be present.

**Safety:** The lithium cell is worn against the body. Use a protected cell with a fuse, never charge while worn, do not charge above 45 °C or below 0 °C, and stop use if the badge becomes warm or swollen or has been crushed. In full sun above about 42 °C ambient the shell can pass 60 °C (DBG-CAL-001, section C), above the cell's and the sensor's limits; wear it shaded or under a vest flap whenever the ambient is above 40 °C in full sun. Do not leave a first build charging unattended.

**Safety:** Exposure logs are health-related personal data. Keep them with the worker, and never use them to discipline workers.

## ■ Open questions

- Accuracy (R3): the error terms in DBG-CAL-001 section F are assumed, and a factor shared across badges misses  $\pm 25\%$  even with the larger reference. Measuring them is TRL 4 work, on hold.
- Range (R2): the sensor's response above  $1 \text{ mg/m}^3$  to mineral dust is unknown and cannot be checked on CalRig as designed.
- Humidity and water spray (R11): the size of the bias from wet cutting and dust suppression sprays is unknown; the badge can flag high humidity.
- Inlet: does the screened downward inlet change sampling efficiency for respirable particles when the worker moves?
- First co-design partner (DBG-DDR-001, O1) and how a site silica fraction is obtained where no laboratory is near (O2). Proposed, awaiting Amish.

Prototype build plan: [DBG-BLD-001](#); design decisions: [DBG-DEC-001](#). Concept media: [blueprint sheet](#), [interactive 3D model](#). General arrangement: [DBG-DWG-001](#). Model: `cad/src/model.py` , exports in `cad/step/` and `cad/stl/` .